Debugging Strategies

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Debugging Strategies before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Loops and Stopping

Without opening the old lesson, explain one key idea from Loops and Stopping. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Debugging Strategies trace and record the first step you can explain without help.

Use W2: Find a revealing case for Testing and **debug**ging functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Debugging Strategies explanation against the supplied facts before

> discussing it.

Use the worked case for Testing and **debug**ging functions. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Test a new function independently

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Debugging Strategies target to a trace and a justified conclusion.

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Debugging Strategies case and show the deciding evidence.

Award two points per matching trait. A bug awards two for every trait.

Test one match and one nonmatch.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A loop should run four times but runs five. You change its colour and stopping condition at once. Which single change is relevant to test first?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 6 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Debugging Strategies adds to Transparent Book Recommender and keep the evidence.

When a ranking surprises you, inspect feature values, weights, arithmetic, and sorting separately.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Intended score is 3 per matching trait. Bug: score starts at 3 and adds 3 for each match. Give expected and actual results for 0,1,2 matches and repair the cause.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Write expectations

> Write expectations. Apply the supplied Debugging Strategies facts and keep your reasoning.

Give expected results for remaining(6, 2), remaining(6, 6) and remaining(6, 0).

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Find a revealing case

Find a revealing case. Apply the supplied Debugging Strategies facts and keep your reasoning.

A faulty function always returns 4. Which of your three cases pass and fail?

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair and retest

Repair and retest. Apply the supplied Debugging Strategies facts and keep your reasoning.

A faulty function adds capacity and used. For (6, 2), state actual and expected. Repair it and name a **regression** case.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Test a new function independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Test a new function independently. Apply the supplied Debugging Strategies facts and keep your reasoning.

Define stock(packs, loose) to return packs * 4 + loose. Create expected tests for (2, 3), (0, 3), (2, 0), then show which detects a faulty packs + loose body.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit the conclusion

Limit the conclusion. Apply the supplied Debugging Strategies facts and keep your reasoning.

All three stock tests pass. Does that prove correct handling of negative counts?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

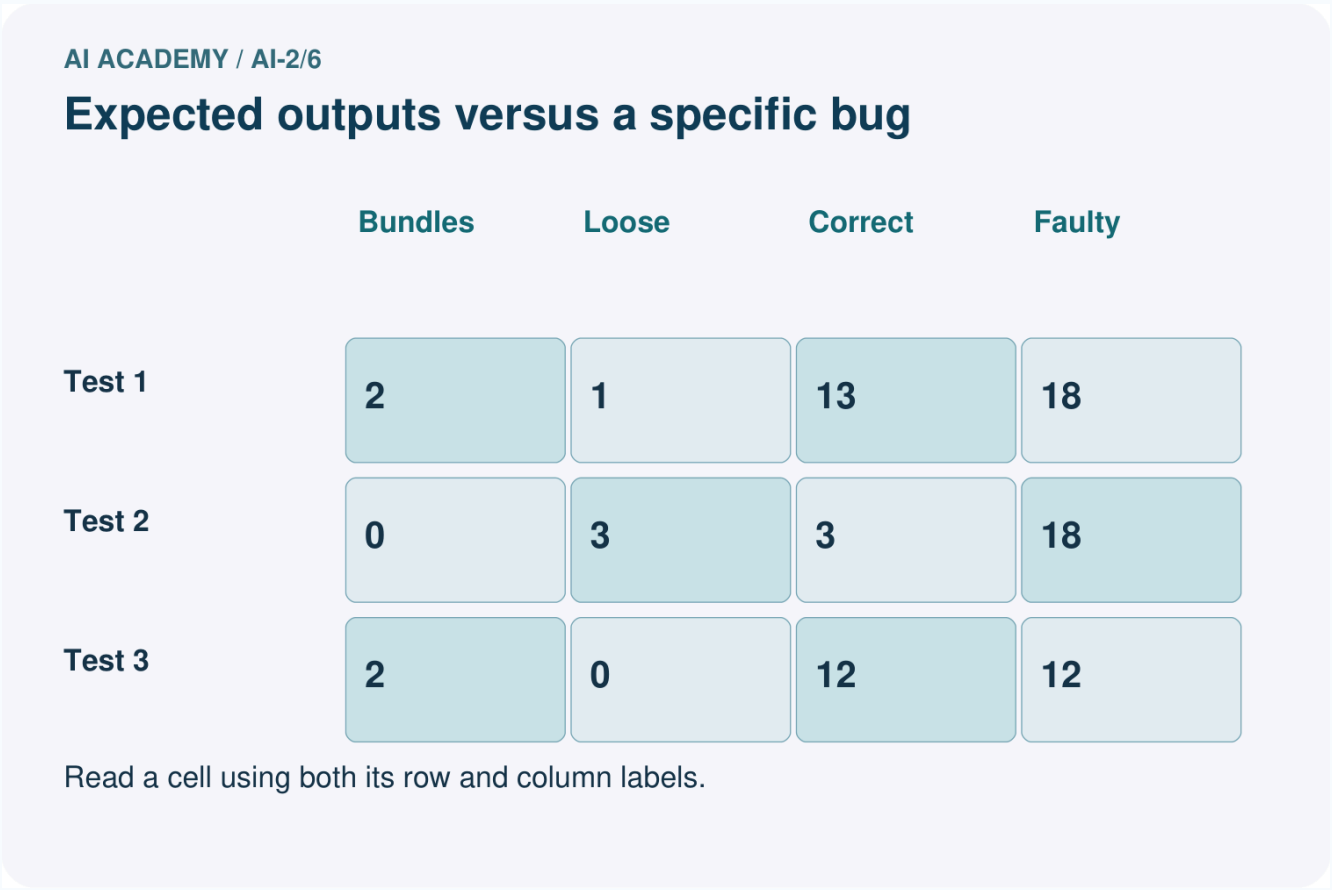
Pictorial case practice: Debugging Strategies

> A fictional shelf function should count books in complete bundles plus loose books. Explain the labelled step before checking the changed condition.

A fictional shelf function should count books in complete bundles plus loose books.

Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose.

Faulty copied body returns (bundle_count+loose)*6. Predict tests before running.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/6

Debugging Strategies

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Expected normal test

For (2,1), correct result is 2 x 6+1=13.



2. Faulty trace

Copied code gives (2+1) x 6=18.



3. Zero-bundle test

For (0,3), correct 3; copied code 18.



4. Nonrevealing test

For (2,0), both return 12; a pass does not locate every bug.

CHANGE ONE THING

Repair the body to bundle_count*6+loose.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which test most directly isolates the handling of loose books, and what should it produce?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Test the recommender's helper functions with normal, zero and **boundary input**s before **debug**ging the whole application.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Test the recommender's helper functions with normal, zero and **boundary input**s before **debug**ging the whole application.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional shelf function should count books in complete bundles plus loose books.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For (2,0), both return 12; a pass does not locate every bug.

1. Zero-bundle test

2. Expected normal test

3. Faulty trace

4. Nonrevealing test

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Nonrevealing test For (2,0), both return 12; a pass does not locate every bug. **Test case**s need not all expose the same error.

Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Repair the body to bundle_count*6+loose. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Expected results for (2,1),(0,3),(2,0) are 13,3,12.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Expected results for (2,1),(0,3),

(2,0) are 13,3,12. Only bundles are groups of six; loose books are counted individually.

Repair a missing building block

Review Loops and Stopping (ai-2/5). Goal: Trace variable values before and after each loop iteration. Short practice: Why is report 0 correct for the original loop, and why is equality-only stopping risky here?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short..

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

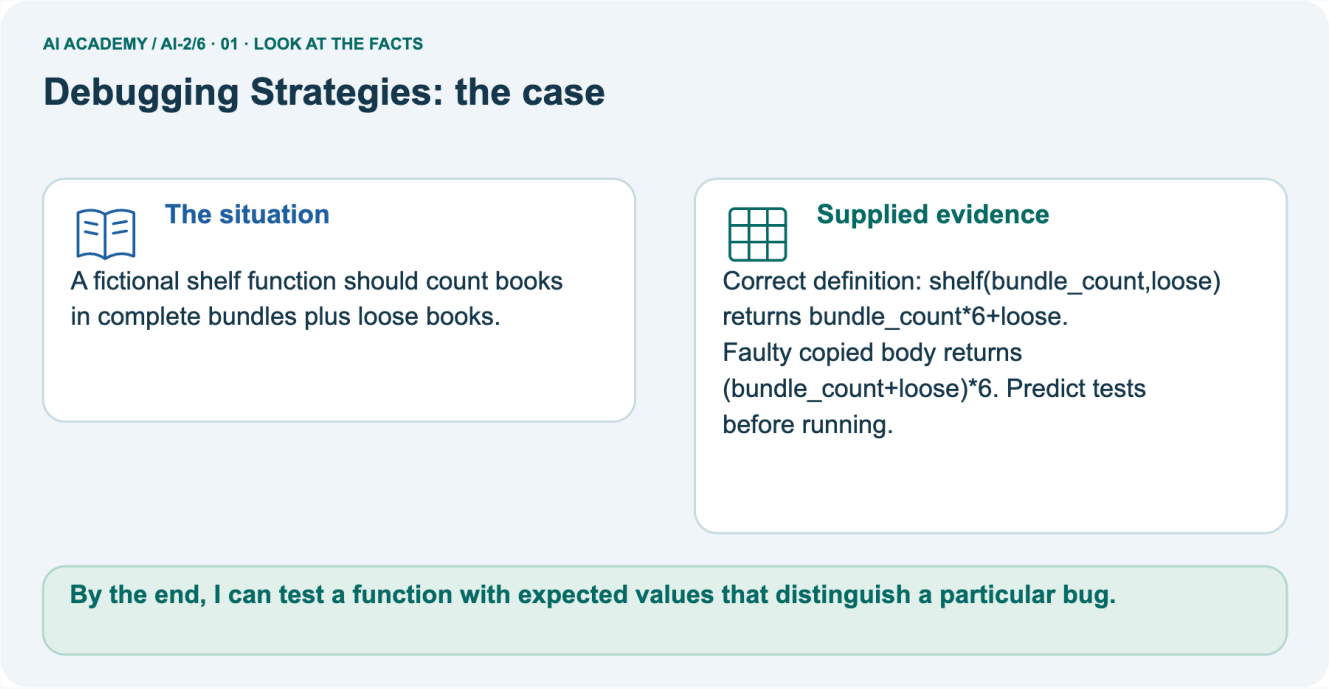
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Debugging Strategies picture from its labels, then record a relevant limit.

01. Debugging Strategies: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

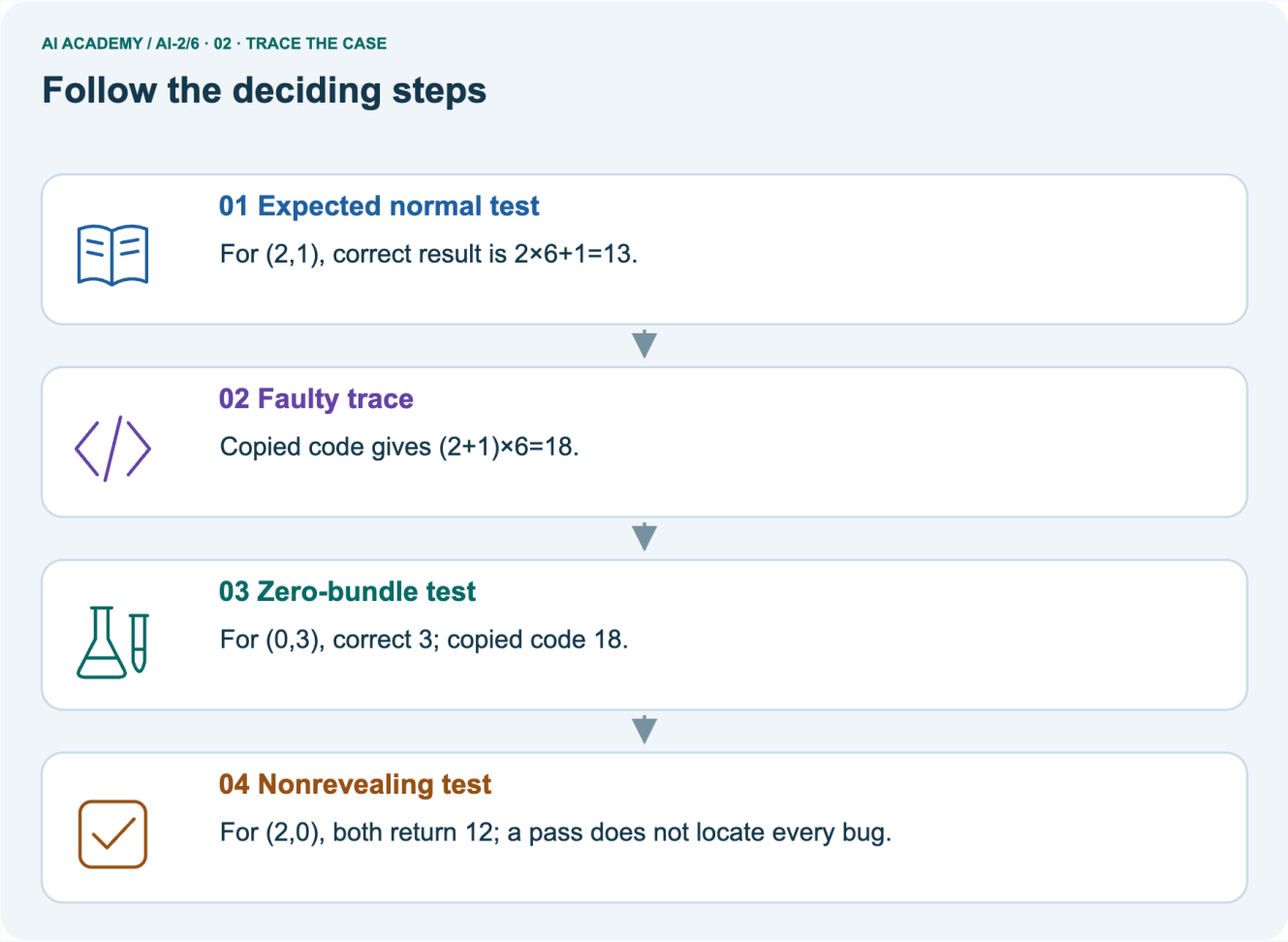
A fictional shelf function should count books in complete bundles plus loose books. Supplied facts: Correct definition: shelf(bundle_count,loose) returns bundle_count*6+loose. Faulty copied body returns (bundle_count+loose)*6.

Predict tests before running.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



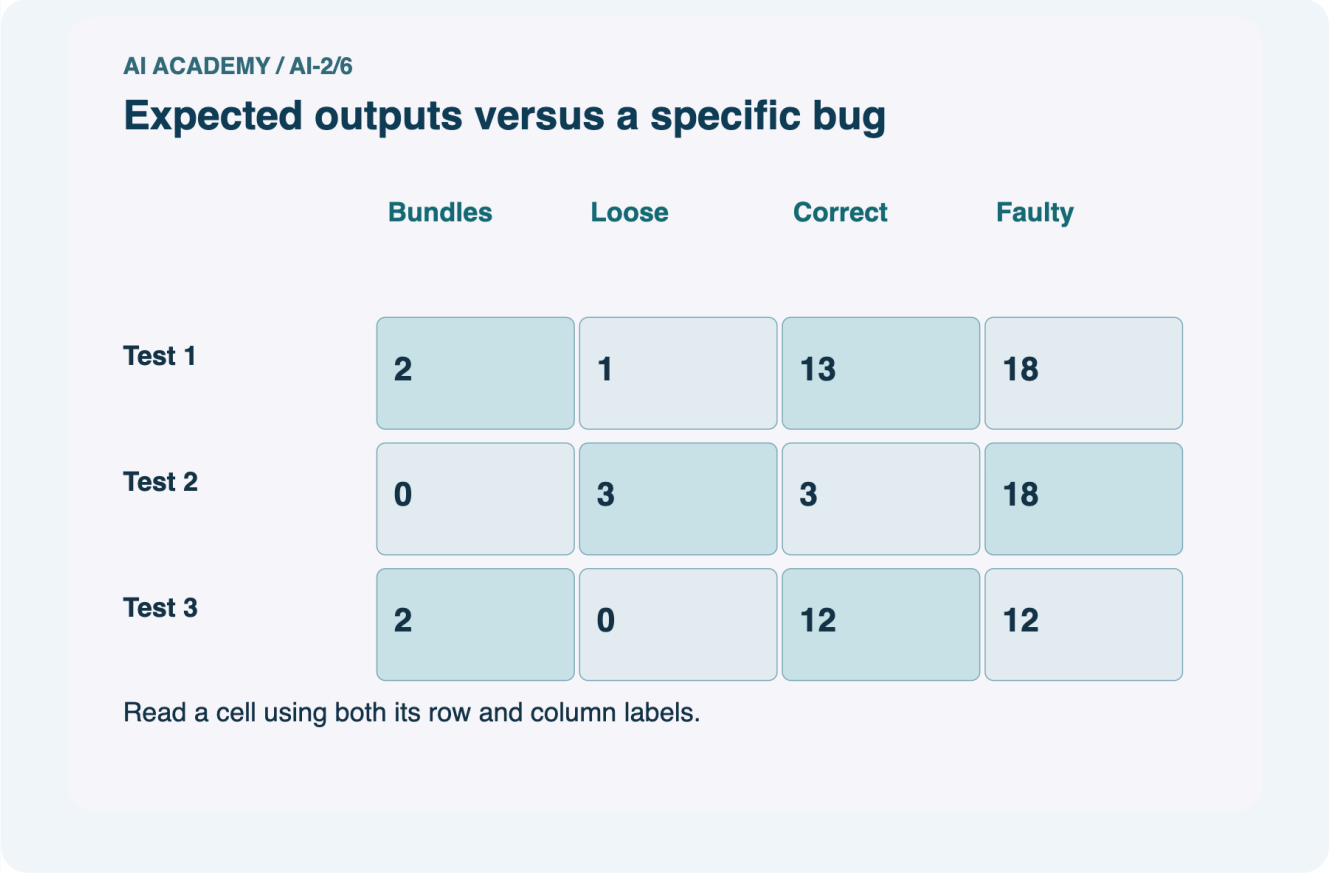
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected normal test: For (2,1), correct result is 2×6+1=13. → Faulty trace: Copied code gives (2+1)×6=18. → Zero-bundle test: For (0,3), correct 3; copied code 18. → Nonrevealing test: For (2,0), both return 12; a pass does not locate every bug.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Expected outputs versus a specific bug



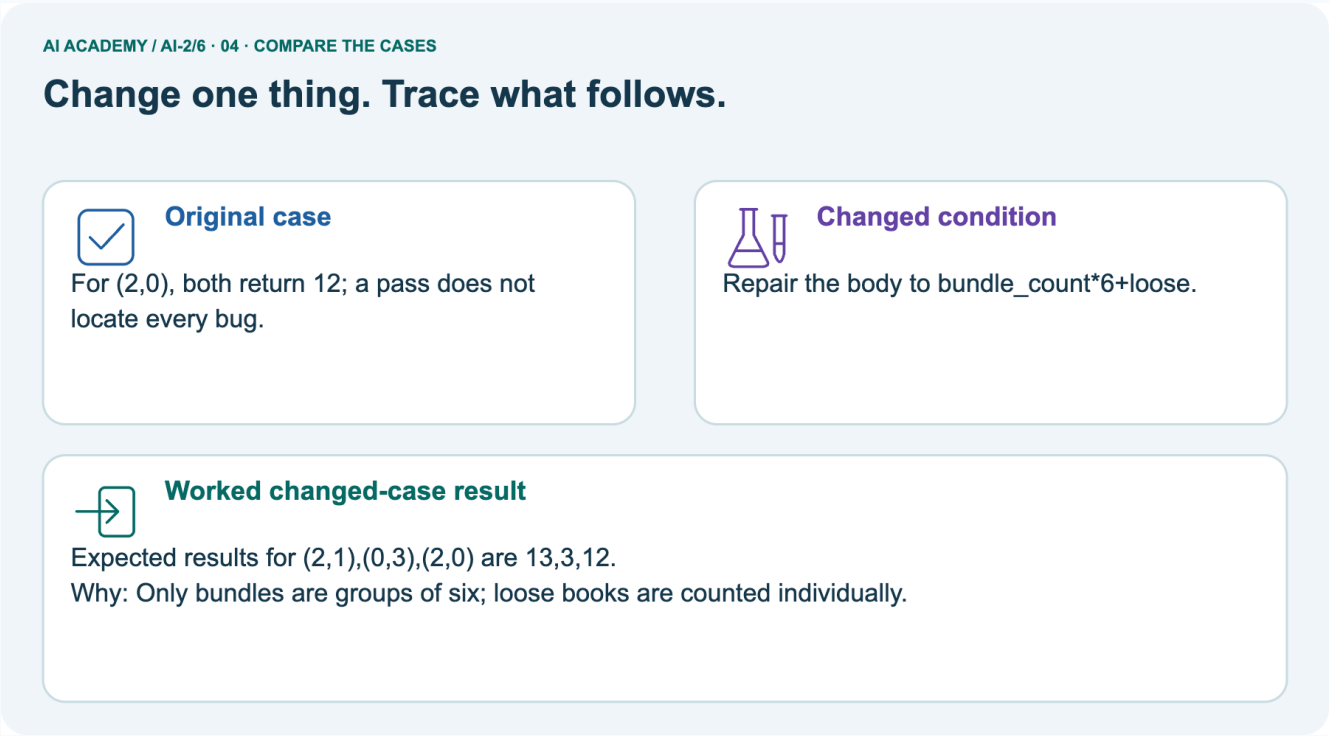
Follow the labelled picture with the matching explanation. Open the vector diagram

Expected **output**s versus a specific bug. Test 1: Bundles = 2, Loose = 1, Correct = 13, Faulty = 18. Test 2: Bundles = 0, Loose = 3, Correct = 3, Faulty = 18. Test 3: Bundles = 2, Loose = 0, Correct = 12, Faulty = 12.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

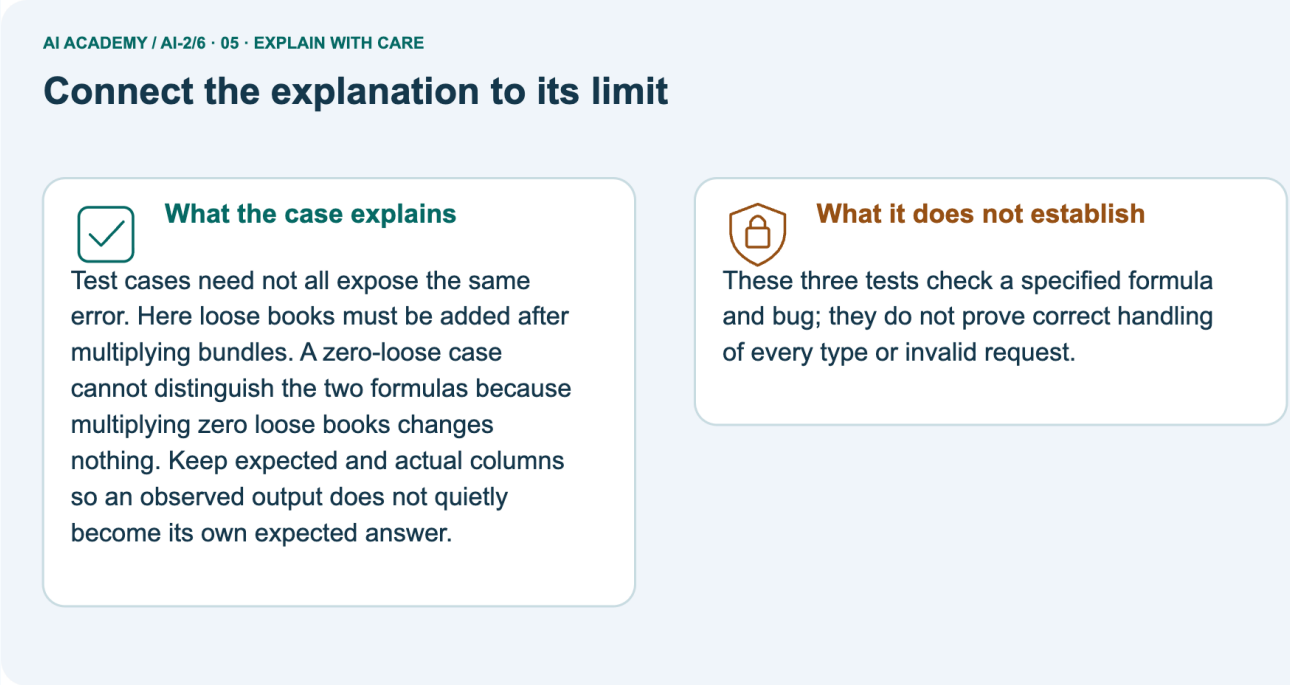
Original case: For (2,0), both return 12; a pass does not locate every bug. Changed condition: Repair the body to bundle_count*6+loose. Expected result: Expected results for (2,1),(0,3),(2,0) are 13,3,12.

Reason: Only bundles are groups of six; loose books are counted individually.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Test cases need not all expose the same error. Here loose books must be added after multiplying bundles. A zero-loose case cannot distinguish the two formulas because multiplying zero loose books changes nothing.

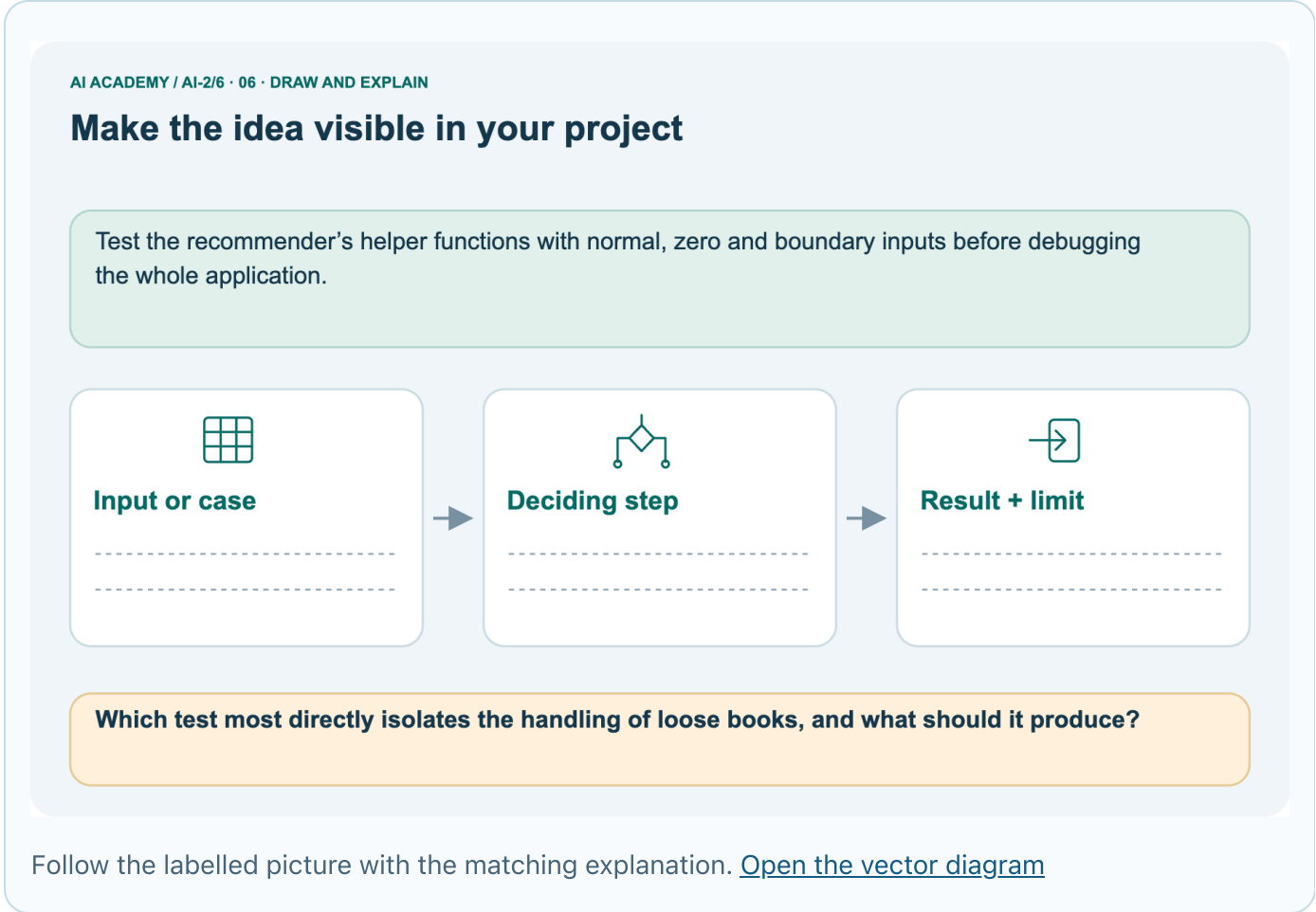
Keep expected and actual columns so an observed **output** does not quietly become its own expected answer.

Limit: These three tests check a specified formula and bug; they do not prove correct handling of every type or invalid request.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test the recommender's helper functions with normal, zero and **boundary input**s before **debug**ging the whole application. Draw your **input**, deciding step, result and limit.

Supported question: Which test most directly isolates the handling of loose books, and what should it produce?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.